⇑Sagar Panchal
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About

ENGINEER ROOTED IN AUTOMATION, MACHINE SOFTWARE, AND INDUSTRIAL SYSTEMS WITH EXPERIENCE ACROSS MANUFACTURING, ROBOTICS AND EMBEDDED TECHNOLOGIES. I BRING SYSTEM-LEVEL THINKING, I VALUE CLARITY AND CONTINUOUS IMPROVEMENT. I AM SEEKING ROLES THAT OFFERS STRONGER OWNERSHIP AND OPPORTUNITY TO DESIGN, BUILD OR ARCHITECT IMPACTFUL ENGINEERING SOLUTIONS - WHETHER IN SOFTWARE, ROBOTICS OR SYSTEMS-ARCHITECTURE.

Education

MS, Electrical & Computer engineering (3.41/4) Tandon School of Engineering, New York University NYU	Dec-17
BE, Instrumentation & Controls, (7.04/10) SVIT, Gujarat Technological University	Jun-14

Skills

Computer Programming:	c/c++/cs, java (desktop & web), git, Python (AI-Automation tools, limited hands-on coding), web(html, css, js), desktop(JavaFx, awt)
Automation:	micro-controllers & single board computers, PLCs(ladder-logic, FBD), hmi/scada
Industries served:	machineries, manufacturing, control panels, solar energy, Education

Work Experience

Software Engineer III - Automation, Universal Instruments, NY — *Sep 22 - present*

- Developing and maintaining **Fuzion** platform software on regular basis.
- Engineered **SBS (Side Barcode Scanning)** using Keyence SR-X300 with Live View, traceability, error recovery and multi-bank support for Connectors with barcodes on side of the body.
- Upgraded existing **AMS** process from single point to 13-locations for higher accuracy on wafer.
- Built a **wafer-to-board mapping** system that converts customer wafer map formats into Universal Fuzion board template files for direct use in production.
- Developed **Selective Placement** with parts buffering, DFF-based Patch & Interposer categories, and avoiding-priority rules for optimal patch-interposer matching.
- Implemented **AVP** to allow user-defined fiducial finds during ECS, NPI and Production flows.
- Integrated Cognex QuickBuild (VPP) with **MCOS↔QBIT** TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Fuzion Precision-Bonder platform.
- Built a CS **Light Intensity Controller** software with multi-channel control, network connectivity and controller discovery for illumination management.
- Implemented **nested-tray** support for Multi-Rot feeders.
- Added **isosceles triangle** fiducial support.
- Implemented configurable **park-position** on Fuzion platform.
- Tech-stack : · CS · C · CPP · Git · VxWorks · Sourcetree.

Automation Engineer , Nor-Cal controls, Ca — *Jan - Jul 22*

- Developed automation software for utility-scale solar power plants
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

Software Engineer - Machine Automation, Mini-Circuits, NY — *Feb 18 - Nov 21*

- Developed and maintained software for pick-and-place rotary machines with computer vision inspection.
- Implemented Reverse motion on rotary machines with Pressing-arm to enable EVT testing.
- Implemented the new shopping cart system for the online store https://www.minicircuits.com/
- Developed enterprise application software for internal engineering teams.
- Tech stack: Java · JavaFX · HTML · CSS · JavaScript

Developer Intern, RevMax Fleet Optimization, NY — *Sep - Dec 17*

- Developed firmware for the SidewalkIQ sensor using Doppler radar to monitor sidewalk activity.
- Deployed sensors at multiple NYC locations and validated field performance
- Conducted R&D to improve data consistency and optimize storage and transfer to backend servers.
- Tech stack : · Arduino-C ·

Instructor (ITEST Program), robotics & mechatronics dpt., NYU — *Sep - Dec 17*

- Mentored to Robotics Club Programming Team for competitions (First Robotics & Vex Robotics).
- Taught concepts of Computer Programming to Robotics Club Team.

Lead Instructor (CREST Program), Center for K12 STEM Education, NYU — *May - Aug 17*

- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
  -Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
  -Wireless Robot Controller - Arduino: https://github.com/sap586/WirelessRobotDay5Activity

Design Engineer, Prima Automation, IN — *Jun 14 - Jan 16*

- Programmed Siemens PLCs in TIA Portal and developed HMI interfaces for industrial automation.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.